

Quick Information:

Description:

This manual walks you through the Ideation Space Robotics Projector system, tells you the basic information needed, walks you through the controls, and some basic troubleshooting.

GitHub:

<https://github.com/RBASOE-Class-Projects/2025-26-TeamLaser>

Website IP:

http://10.40.97.7/

Wifi Name:

UIndyEngineering

Set Up:

To set up a new Raspberry Pi:

Ansible is used to automate the setup for a new raspberry pi

Running the ansible code will setup the pi

- `sudo apt install ansible`
- clone or ssh the files onto pi
- `cd 2025-26-TeamLaser/ansible`
- `ansible-playbook -i inventory.ini site.yml --ask-become-pass -k`
- `sudo reboot`

This will set up the system on a fresh raspberry pi, and rebooting will ensure the system boots to the correct ip and the backend starts.

To access the website:

Connect to the UindyEngineering Wifi

Go to <http://10.40.97.7/>

User Information:

Website Controls Explained

— Home Tab —

Ideation Floor Projector:

Select image:

To use

- Click the dropdown to be able to select the racetrack you wish to project

Image scale:

To use

- Can slide the slider to a certain value you wish. 100% is the original size of the selected image. With 50% making the selected image half the size it was.
- Or can input an exact value in the textbox below the slider.

Projection:

To use

- Click the project button to project the selected image, with all configurations to it (Image scale and keystone calibrator).

Stopwatch:

To use

- Click start to start the stopwatch.
- Click stop to pause the stopwatch. You can also click start again to continue the stopwatch.
- Click reset to bring the stopwatch to its original starting at 00:00:00.00.

Projector control:

To use

- Press "Projector ON" to turn on the projector.
- Press "Projector OFF" to turn off the projector.
- The status is used for troubleshooting to see if the program detects the projector is on or off.

Boundary Detection Control

To use

- This is a work in progress right now.

— Settings Tab —

System Lock:

Status

- “System Unlocked” - This means everyone on the website can click every button.
- “System Locked” - This means the tab that entered the admin password is the only one who can click buttons.

To use

- Locking - Input the admin password, then click “Lock System”.
- Unlocking - Input the admin password, then click “Unlock System”.

Upload Image:

To use

- Click “Choose File” to upload an image file from your computer. The support image formats are: .png, .jpeg, and .jpg.
- Enter in the upload image password, and click “Upload New Image”
- The image will be in the database, and can be used to project after five seconds. The name of the image will be based on what you name the image in your computer files.

Delete Image:

To use

- Click the dropdown menu called, “Select an Image...” and click on the name you wish to delete.
- Then upload the delete image password, and click “Delete Image”

System Reboot:

To use

- In order to use this, you must enter the admin password in the “System Lock” setting at the top of the settings tab.
- Once the admin password is entered, then you can click “Reboot Raspberry Pi” it will then reboot the Raspberry Pi and the software will automatically start up again.

Keystone Calibrator:

To use

- Click “Show Keystone Calibrator” to bring up more controls.
- You can drag the red dots on the image to adjust the skew to the projected image.
- You can press “Save Keystone” in order to keep this calibration for the selected image for later times to project.
- You can press “Reset” to reset the selected image to the defaulted image skew.
- You can press “Enable Live” in order to edit the keystone calibration, with the changes being shown live. You will drag the corners with your mouse and it will flash on the projection and change. You can press the “Escape” key on your keyboard to leave the live keystone calibrator.

Troubleshooting

System is not turning on:

1. On the control website, press turn off, and then turn on. If the website is down, please skip this step.
2. Unplug Raspberry Pi and plug it back in. To tell if the Raspberry Pi is on, there should be a bright green light on.
3. If the Raspberry Pi is not turning on, then look up manuals on how to fix said Raspberry Pi online.

Projection is wonky looking:

1. This issue is fixed by going into the keystone calibration menu, and pressing reset and save, then projecting again.

Website is not loading:

1. Make sure you are login to the Uindy Engineering Wifi
2. Make sure the website address is correct: <http://10.40.97.7/>
3. Make sure the Raspberry Pi is on. (You will see a green light shining on the projector mount)
4. Look at console logs on the website, to look for errors in order to fix

Buttons are unresponsive:

1. Go to the settings tab, and look at the system lock section. If it says "System Lock" you need to or the admin needs to enter the admin password to unlock the system.